Piecewise function

$$f(x) = \begin{cases} \dfrac{x^2+4x}{x+4} & \text{if } x < -2 \\ x+2 & \text{if } -2 \le x \le 2 \\ -2 & \text{if } x > 2 \end{cases}$$

these are the condition "x" must follow to use the equation

$\frac{x^2+4x}{x+4}$ | $x+2$ | $-2$

$x < -2$ | $-2$ | $-2 \le x \le 2$ | $2$ | $x > 2$

created number line and placed the condition where they belong with there equation

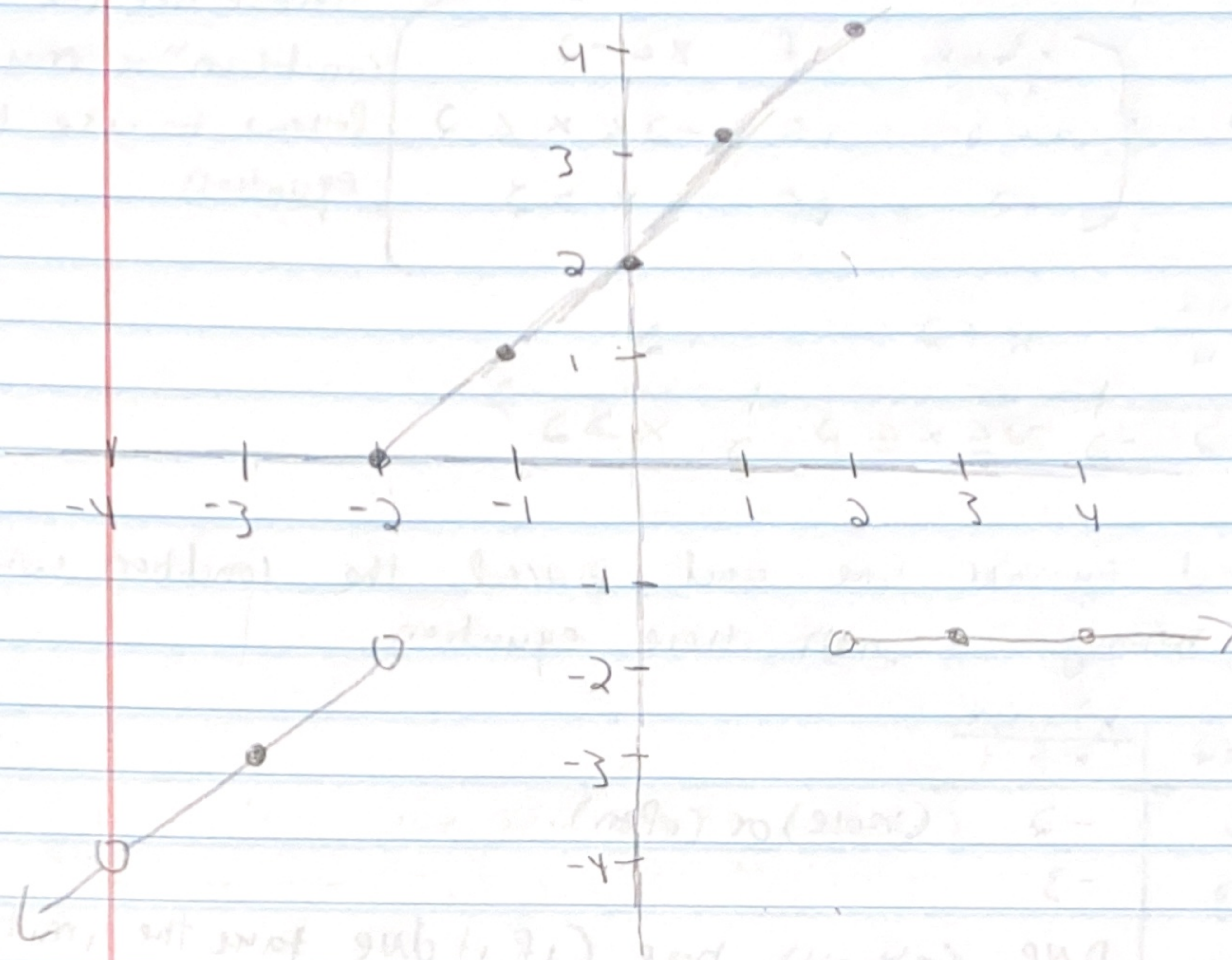
| $x < -2$ | $\frac{x^2+4x}{x+4}$ |
|---|---|
| $-2$ | $-2$ (hole) or (open) |
| $-3$ | $-3$ |
| $-4$ | DNE $(-4,-4)$ hole (if it dne take the limit) |

$$\lim_{x \to -4} f(x) \; \frac{x^2+4x}{x+4} = \frac{x\cancel{(x+4)}}{\cancel{(x+4)}} = -4$$

| $-2 \le x \le 2$ | $x+2$ |
|---|---|
| $-2$ | $0$ |
| $-1$ | $1$ |
| $0$ | $2$ |
| $1$ | $3$ |
| $2$ | $4$ |

| $x > 2$ | $-2$ |
|---|---|
| $2$ | $-2$ (hole) or (open) |
| $3$ | $-2$ |
| $4$ | $-2$ |

for $x < -2$ say that x must be less then two but for our first x-value we had a a $-2$ we did this because we want to know where the hole will be do this for everyone bc it says you can do it it will be closed like $-2 \le x \le 2$ but if it say you can't it has to be open




$x < -2$ can keep going for infinite

$x > 2$ can keep going for infinite

$-2 \leq x \leq 2$ is restricted from -2 to 2

$$\lim_{x \to 4} \sqrt{\frac{x-4}{x^2-4x}}$$

$$\sqrt{\lim_{x \to 4} \frac{x-4}{x^2-4x}} \longrightarrow \sqrt{\lim_{x \to 4} \frac{\cancel{(x-4)}}{x\cancel{(x-4)}}} = \sqrt{\lim_{x \to 4} \frac{1}{4}}$$

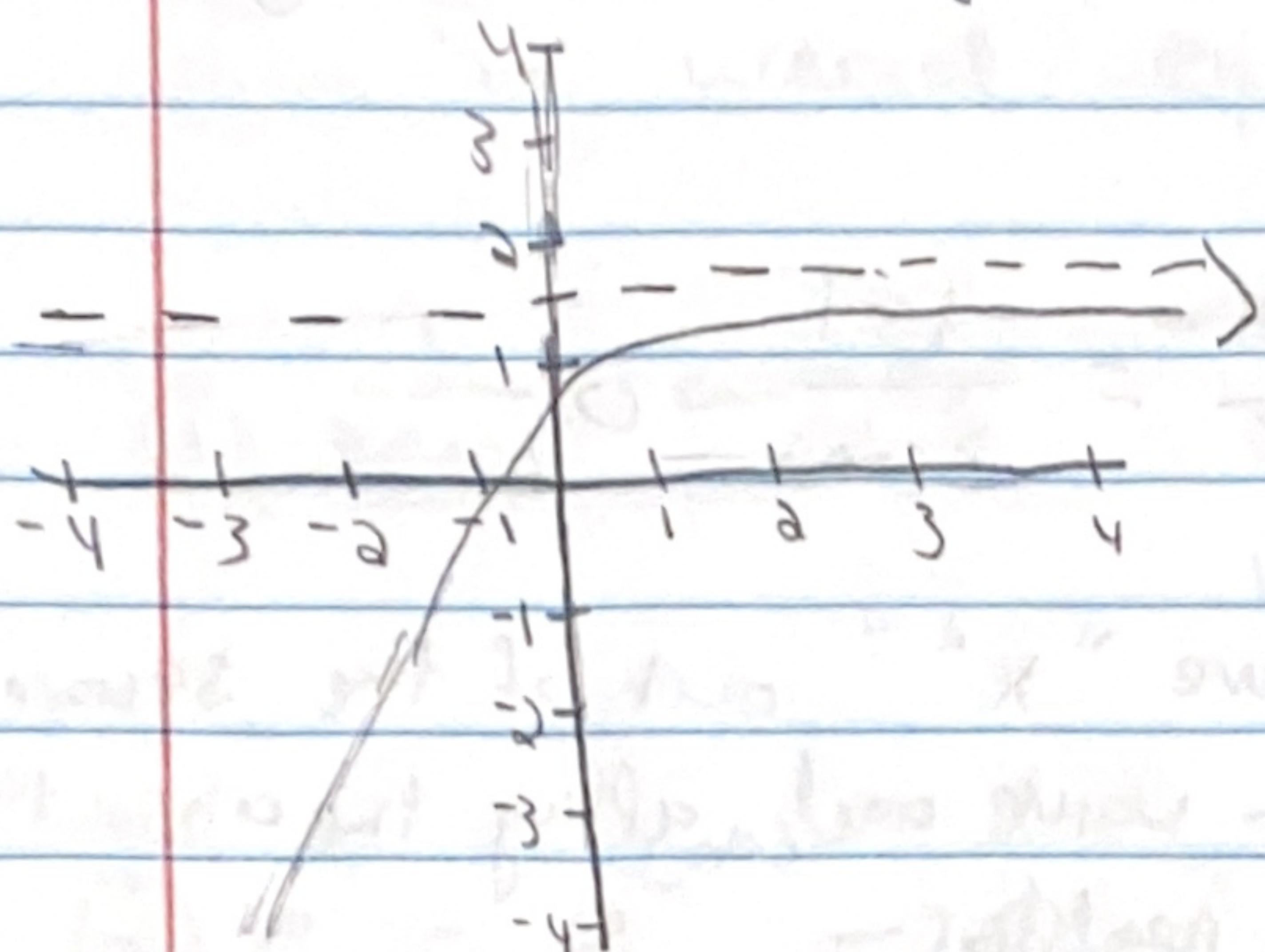
$$\sqrt{\frac{1}{4}} = \boxed{\frac{1}{2}}$$

$$\lim_{x \to 3} LN\left(\frac{x^2-1}{x-1}\right) \longrightarrow LN\left(\lim_{x \to 3} \frac{x^2-1}{x-1}\right)$$

$$LN\left(\lim_{x \to 3} \frac{(3)^2-1}{3-1}\right) \longrightarrow LN\left(\frac{8}{2}\right)$$

# horizontal asymtote and vertical asymtote

this is an graph example of a horizontal line



the dotted line is the horizontal asymtote

it must be call $y =$

to find the horizontal asymtote of any function you must take "x" to infinite

example: $\frac{4x^2 - 8x + 3}{1 - 4x^2}$

$\lim_{x \to \infty} \frac{4x^2 - 8x + 3}{1 - 4x^2}$

$\lim_{x \to \infty} \frac{4\cancel{x^2}}{-4\cancel{x^2}}$

then pick the term with the highest degree on the numerator and dunominator

this only work when $x \to \pm\infty$

$\boxed{\lim_{x \to \infty} = -1}$ $\boxed{y = -1}$

$\lim_{x \to -\infty} \frac{4x^2 - 8x + 3}{1 - 4x^2}$

also take the equation to negative infinite and find the term with the highest degree on the numerator and dunominator

$\lim_{x \to -\infty} \frac{4\cancel{x^2}}{-4\cancel{x^2}}$

$\boxed{\lim_{x \to -\infty} = -1}$ $\boxed{y = -1}$

☆ they dont have to be the same answer ☆

example # 2

find the horizontal asymptote

$$f(x) = \frac{3-\sqrt{25x^2+1}}{5x+3x^2}$$

$$\lim_{x \to \infty} \frac{-\sqrt{25x^2}}{3x^2} = \frac{-15x1}{3x^{\cancel{2}}} = \frac{-151}{3(\infty)} = 0$$

remember when your remove "$x^2$" out of the square root to add a absolute value and apply the absolute value definition when you need to

$$\boxed{y=0}$$

$$\lim_{x \to -\infty} \frac{-\sqrt{25x^2}}{3x^2} = \frac{-15x1}{3x^{\cancel{2}}} \quad \frac{-5}{3(-\infty)} = 0$$

they do not need to have the same answer they can have different ones

example #3

$$f(x) = \frac{4x^2-8x+3}{1-4x^2} =$$

your answer has to be repersented as

$$\boxed{y=-1}$$

$$\lim_{x \to \infty} \frac{4\cancel{x^2}}{-4\cancel{x^2}} = -1$$

$$\lim_{x \to -\infty} \frac{4\cancel{x^2}}{-4x^2} = -1$$

the horizantal asymptote is when x is going to either positive infinite or negitive infinite and its asking for the y-value or f(x) when you have to take the limit to infinity (Positive or negitive) and you come to weird equation use this to help

① $\frac{1}{3(\infty)^2} \to 0$ - ether for positive or negitive infinite you will still get a zero

② $e^{-x} = e^{-(\infty)} \to 0$ - either for positive or negitive infinite you will still get a zero

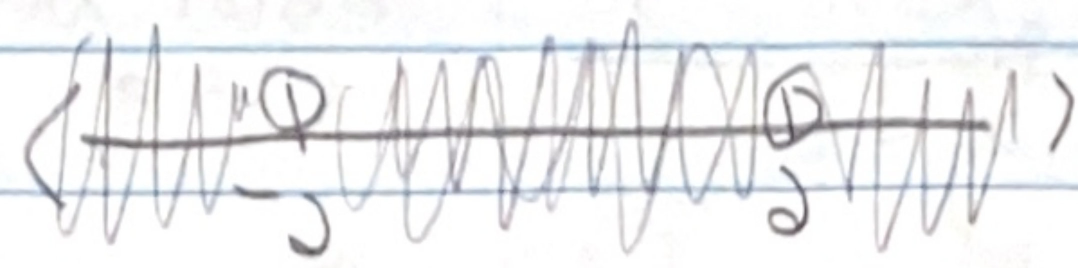
③ $\sqrt{\infty} = \infty$ - a negitive infinite in a square root is undefined

④ $x(\infty)$ - x being any number will just be "$\infty$" if the number was a negitive the answer will be "$-\infty$"

⑤ $\frac{\infty}{6}$ or $6 \cdot \infty$ - Positive infinite divide or multiplyed by and number is just "$\infty$" same thing with a negitive multiplication or division it will be a "$-\infty$"

vertical asymtote, Domain, holes

$f(x) = \frac{x+2}{x^2-4}$




Domain: $x^2-4 \neq 0$ $\boxed{(-\infty,-2) \cup (2,\infty)}$

$x \neq \pm 2$

this can only be used when stating the domain when stating interval of continuity use commas

- to figure out the VA and holes you must take the limit of the domain

$\lim_{x\to -2} f(x) = \frac{x+2}{x^2-4} \to \frac{\cancel{(x+2)}}{(x-2)\cancel{(x+2)}} \to \frac{1}{-2-2} = -\frac{1}{4}$

we know that $f(-2) = DNE$ and we also know that the $\lim_{x\to -2} f(x) = -\frac{1}{4}$ these two are not the same which mean that this is our hole

$\boxed{(-2, -\frac{1}{4})}$

$\lim_{x\to 2} f(x) = \frac{x+2}{x^2-4} \to \frac{\cancel{(x+2)}}{(x-2)\cancel{(x+2)}} = \frac{1}{2-2} = \frac{1}{0}$ cannot apply limit laws

$\boxed{VA: x=2}$

this is our VA this can only be done if there are two number in domain (example Next Page)
You must simplify before you apply limit

$\lim_{x\to \infty} f(x) = \frac{x+2}{x^2-4} = \frac{\cancel{x}}{x^{\cancel{2}}} = \frac{1}{\infty} = 0$

$\boxed{HA: y=0}$

$\lim_{x\to -\infty} f(x) = \frac{x+2}{x^2-4} = \frac{\cancel{x}}{x^{\cancel{2}}} = \frac{1}{-\infty} = 0$

$(-1)^2 = 1$

$f(x) = \dfrac{2x - \sqrt{3x^2+1}}{x-1}$

anything that goes in the "$x^2$" either negitive or possitive will be fine

$-1$

Doman: $x - 1 \neq 0$
$x \neq 1$

$\boxed{(-\infty, +1) \cup (+1, \infty)}$

again to find the holes and VA you must find the limit of the domain

$\lim_{x \to 1} f(x) = \dfrac{2x - \sqrt{3x^2+1}}{x-1} \cdot \dfrac{2x + \sqrt{3x^2+1}}{2x + \sqrt{3x^2+1}}$

$a^2 - b^2 = (a-b)(a+b)$

$\dfrac{(2x)^2 - (\sqrt{3x^2+1})^2}{(x-1)(2x+\sqrt{3x^2+1})} = \dfrac{4x^2 - (3x^2+1)}{(x-1)(2x+\sqrt{3x^2+1})} \to \dfrac{4x^2 - 3x^2 - 1}{(x-1)(2x+\sqrt{3x^2+1})}$

$\dfrac{x^2-1}{(x-1)(2x+\sqrt{3x^2+1})} \to \dfrac{\cancel{(x-1)}(x+1)}{\cancel{(x-1)}(2x+\sqrt{3x^2+1})} \to \dfrac{1+1}{(2(1)+\sqrt{3(1)^2+1})}$

$\dfrac{2}{2+2} = \dfrac{2}{4}$ $\boxed{\dfrac{1}{2}}$

We know that the lim as x approach 1 is 1/4 and that is not equal to f(1) = DNE which mean the hole is $\boxed{(1, 1/2)}$

VA: We do not have a vertical asymtope for this function we only have one number in the doman and that number turned out to be just a hole not a VA usally you have two number in the domain but for this function you only have one

HA: $\frac{2x-\sqrt{3x^2+1}}{x-1}$

$|x| = \begin{cases} (x) & \text{if } x \geq 0 \\ -(x) & \text{if } x < 0 \end{cases}$

$\lim_{x \to \infty} \frac{2x-\sqrt{3x^2}}{x} = \frac{2x - x\sqrt{3}}{x} = \frac{\cancel{x}(2-\sqrt{3})}{\cancel{x}} = 2-\sqrt{3}$

$\lim_{x \to -\infty} \frac{2x-\sqrt{3x^2}}{x} = \frac{2x - -(x)\sqrt{3}}{x} \to \frac{2x+x\sqrt{3}}{x}$

$\frac{\cancel{x}(2+\sqrt{3})}{\cancel{x}} = 2+\sqrt{3}$ $\boxed{y = 2-\sqrt{3} \text{ and } y = 2+\sqrt{3}}$

the first one "x" is approaching positive inife and the rule say $x \geq 0$ the the absolute value is $(x)$ but for the second one $x$ is approaching negative infinite and the rule say $x < 0$ then the absolute value is $-(x)$

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$\sqrt{x^2} = x$ — what of we said that $x = -1$ does that mean that the "$-1$" came out the square root as $-1$ well

$|x|$

No!

$x = -1$ $-(-1) = 1$

but

$x = 1$ $(1) = 1$

— that is why we give it a absolute value so even if it was a negitive one would be a positive

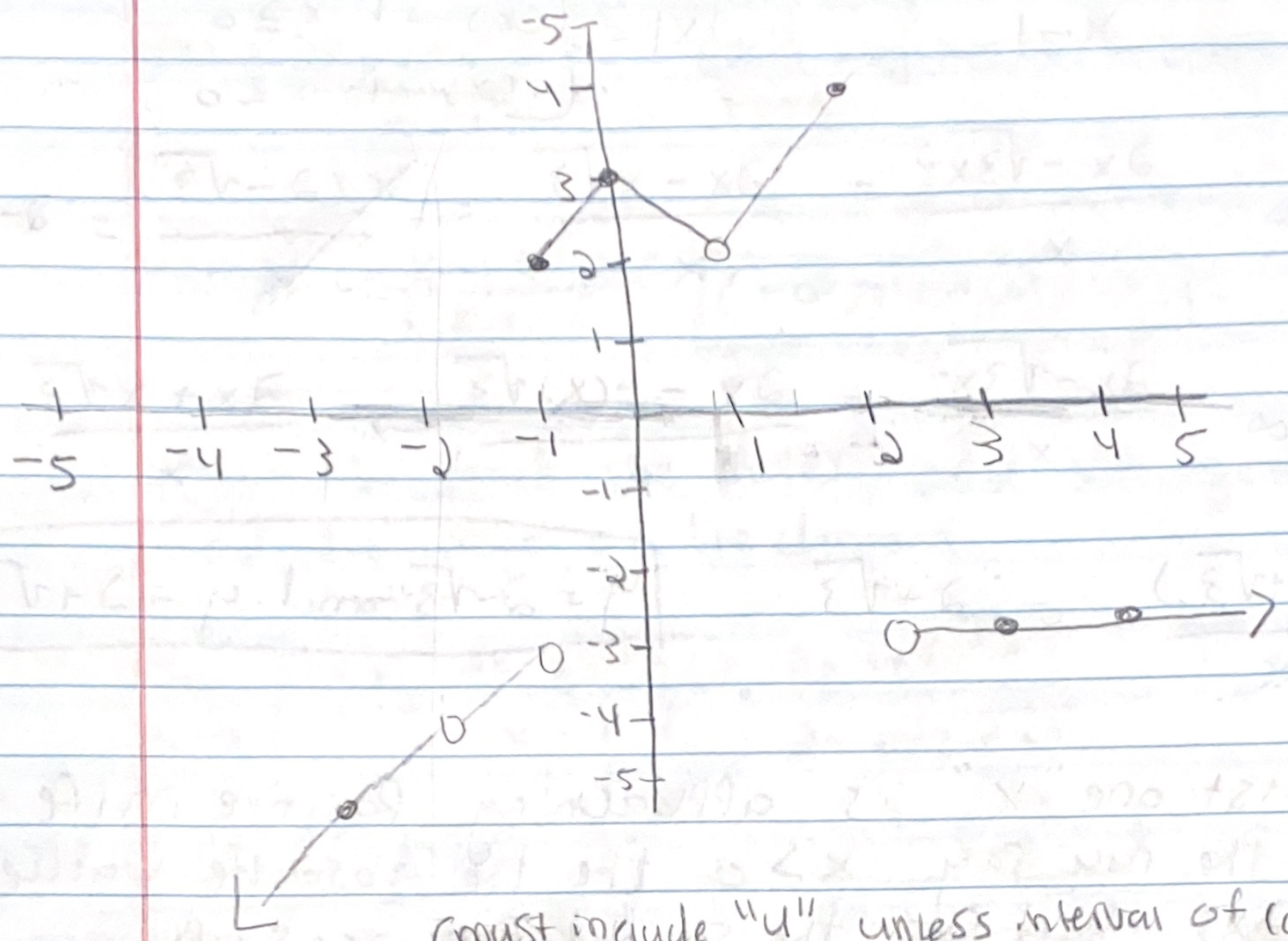
the point is we need to change it to satisfy the absolute value because nothing can come out a square root negitive

by definition the absolute value is

$|x| = \begin{cases} (x) & \text{if } x \geq 0 \\ -(x) & \text{if } x < 0 \end{cases}$ meaning of $x$ is greater then zero we can write is as $(x)$ but if $x$ was less then zero we need to write it like this $-(x)$

if your x-value is going to the negitive including infinit you will use $-(x)$ because of this $x < 0$ if it is going to the positive including infinite you will use this $(x)$ because $x \geq 0$

Doman on graph




(must include "u" unless interval of continuity)

Domain: $(-\infty,-2]\cup(-2,1)\cup(1,\infty)$

the domain is the x-values nothing else beginning from negitive infinite we make our first stop at "-2" why because there isn't any other point that tell use x can equal -2 -1 cannot work because -1 can equal 2 Just because there is a hole under -1 does not mean "-1" cam Join the domain there is a closed dot right on top of it. looking at positive 1 that x-value cannot work because there is only a hole there not a close dot any where else.